

Helping Hand Beta Build

M4 Milestone Work Outline

Course: CEN 4908C

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Repository snapshot: October 2, 2026; beta app commit [d581d19](#)

Helping Hand combines an ESP32 sensor glove with an Android Flutter application for American Sign Language learning. The following work was completed for the beta milestone and is represented in the project repository.

1 Work completed

- **Account access and learner progress.** Added email/password account creation and sign-in, password reset, sign-out, and user-scoped progress handling. Added repository tests for progress isolation between user accounts. The account-flow and isolation-test changes are recorded in commits [4259269](#) and [873b589](#).
- **Learning interface.** Refreshed the application identity and learner screens. Alphabet and Numbers let learners choose a target; Words provides a vocabulary list. These interface changes are recorded in commits [9c5caab](#) and its preceding UI commits.
- **Target-specific practice navigation.** Connected selected letter and number targets to separate practice pages with target labels and Back navigation. Integrated those pages with the app-shell-owned BLE state and retained the static practice feedback path. The route change is recorded in commit [d581d19](#).
- **Milestone documentation.** Updated the project README with the current architecture, setup and build instructions, beta work summary, and known-issue list. Prepared the beta test plan as a separate submission document. The README update is recorded in commit [d807ef8](#).

The beta implementation continues to use the existing static 36-class letter-and-number classifier in practice. The linked repository commits provide timestamped source evidence for the work listed above.